

Grounding Creative Query Suggestions, Faithfully and Efficiently: From Corpus-Grounded Supervision to Reinforcement Learning

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Abstract

Creative search platforms help users turn short, vague queries into searches and generation prompts. To be useful, a suggestion has to fit the context it appears in, the asset type, the season, and the user’s region, and it has to be grounded in assets the platform can actually retrieve. A fluent suggestion that returns nothing wastes the user’s attention and erodes trust. Building such a system is hard because there is no ready-made training data: user logs are sparse, noisy, and privacy-constrained, and they say nothing about context or grounding. We first solve this with corpus-grounded synthetic supervision. A two-stage offline pipeline turns 59M asset captions into context-aware training data, constrained by a Creative Knowledge Graph and grounded through retrieval, and fine-tunes a single low-latency model that serves related suggestions, refinements, and generation-ready prompts in one retrieval-free step. We check grounding by running suggestions against the production index: offline, retrieval grounding raises the share of well-grounded suggestions by over 30% relative, and in production about 99% of served suggestions return real assets (a null rate near 1%). The deployed system is preferred over the prior log-based suggester in human evaluation and improved the primary business metric in production. As the system reached new surfaces, two requirements emerged that the supervised data met poorly: single-word suggestions for space-constrained interfaces, and *selective* seasonal and regional context—added in the harder cases the deployed model missed, and suppressed where it does not apply. Rather than regenerate the data and retrain, we add a Group Relative Policy Optimization (GRPO) stage whose hybrid reward pairs an LLM judge with programmatic checks for format, length, and diversity. It improves seasonal alignment by 18.6 and regional alignment by 5.2 points (on a 0 to 100 judge scale) and raises single-word compliance from 15.1% to 99.1%, a format absent from the

supervised data, while keeping relevance and diversity preserved in the normal multi-word regime and leaving the retrieval-free serving path unchanged.

1 Introduction

On a creative search platform, users start from a short, vague query such as “birthday poster” or “couple on beach” and expect the system to help them turn it into something they can act on. A *query suggestion* system does this by proposing related searches that explore nearby ideas, narrow refinements that add specificity, and generation-ready prompts for creating new assets. Good suggestions help users find content they would not have thought to search for, and they are a major driver of search engagement.

What makes a suggestion useful is not fluency but fit to context, and three signals matter. The *asset type* shapes what a good suggestion looks like: a suggestion for a template is not the same as one for a video, an image, or an icon. *Seasonality* should shape a suggestion when the query calls for it, for example heart motifs near Valentine’s Day, but stay absent when it does not. *Region* should influence cultural styling without ever naming a place. And every suggestion must be *grounded*: it should point to content the platform can actually retrieve or generate. A fluent suggestion that returns no assets wastes the user’s attention and erodes trust.

Traditional query suggestion learns these patterns from user logs and clicks, which does not work well here. Creative queries are sparse, noisy, and long-tailed; logs underrepresent the structured refinements and long prompts we need; and behavioral data carries privacy constraints. Logs also say nothing about the context conditioning above, and give no guarantee that a suggestion maps to a real asset. There is simply no ready-made dataset to learn from.

Prior work (Jain et al., 2026) addresses this

by replacing logs with corpus-grounded synthetic supervision. A two-stage offline pipeline turns 59M asset captions into context-aware training data, constrained by a Creative Knowledge Graph and grounded through retrieval, and trains a single retrieval-free model that serves all three suggestion types. We summarize that system (§3.1) and contribute its reinforcement-learning extension.

The system is deployed and effective, but as it reached new product surfaces, two requirements emerged that it met poorly. First, surfaces such as filters, chips, and autocomplete need *single-word* suggestions, a format the training data never contained. Second, seasonal and regional context has to be applied *selectively*: the deployed model adds it in easy cases but misses the harder ones where it belongs, staying neutral by default. Both are easy to state as goals but hard to add through more data. Single-word output has no examples to learn from, and regenerating and retraining the whole pipeline risks breaking what already works in production.

These behaviors are far easier to *score* than to *demonstrate*: a single-word check is exact, and selective context can be judged case by case. We therefore use reinforcement learning as a lightweight way to adapt the model. Starting from the deployed checkpoint, we add a Group Relative Policy Optimization (GRPO) (Shao et al., 2024) stage that optimizes a hybrid reward: an LLM judge for context quality, together with programmatic checks for format, length, and diversity. GRPO stays anchored to the deployed model, so it adds the new behaviors without eroding the grounding already in place, and because it changes only weights, the single-step, retrieval-free serving path is unchanged.

Contributions.

- A deployed creative query-suggestion system that grounds suggestions in a 59M-caption asset corpus through synthetic supervision, replacing user logs, and checks grounding by running suggestions against the production index (about 99% of served suggestions return real assets in the deployed system).
- A GRPO stage whose hybrid reward adapts the deployed model to new requirements, selective seasonal and regional grounding and single-word generation, without regenerating supervision, changing the serving path, or eroding the grounding already in place.
- Results showing improved seasonal and regional alignment and reliable single-word generation that the supervised data never covered, with relevance and diversity preserved in the normal multi-word regime.

2 Related Work

Grounding and faithfulness. Most work on grounding language models targets *factual* faithfulness: retrieval-augmented generation conditions outputs on retrieved documents to reduce hallucination and improve provenance (Lewis et al., 2020), and a large literature studies when and why generated text departs from its source (Ji et al., 2023). Our setting shares the concern but differs in what a suggestion must be faithful to. A creative query suggestion is not a factual claim to be verified against evidence; it is useful only if it corresponds to assets the platform can actually retrieve or generate. We therefore treat grounding as *corpus* or *inventory* grounding: a suggestion is grounded when executing it against the production index returns relevant assets. This makes grounding directly measurable in deployment (the null rate of §3.1) rather than a proxy for factual correctness, and it is the property Stage 1 establishes and Stage 2 preserves.

Query suggestion and reformulation. Classical query suggestion exploits logs, clicks, and session co-occurrence (Baeza-Yates et al., 2004). Recent work reframes it generatively with LLMs, aligning recommendation with preference signals (Min et al., 2025) and augmenting suggestion with external knowledge (Baek et al., 2024). These approaches remain dependent on behavioral data or personalization signals. ? replace logs entirely with corpus-grounded synthetic supervision; this work adds reinforcement learning on top of that supervised model to meet requirements the supervised data does not cover.

Synthetic supervision for retrieval and generation. Using LLM-generated data in place of human- or log-derived labels is increasingly common for addressing sparsity, privacy, and domain specificity (Bonifacio et al., 2022; Dai et al., 2023; Thakur et al., 2024), including synthetic query generation for retrieval (Nogueira et al., 2019; Jeronimo et al., 2023). We instead adapt a knowledge-graph-constrained, corpus-grounded supervised model with RL rather than regenerating data, trading a retraining cycle for reward design.

RL post-training of LLMs. RLHF (Ouyang et al., 2022) and PPO (Schulman et al., 2017) optimize policies against a learned reward model; DPO (Rafailov et al., 2023) removes the explicit reward model via preference pairs. GRPO (Shao et al., 2024) drops the value network and estimates advantages from group-relative rewards, and underpins reasoning-focused RL (DeepSeek-AI, 2025). We adopt GRPO for two reasons. Its group-relative formulation fits set-valued suggestion outputs and a heterogeneous, partly programmatic reward. And initializing from a deployed supervised checkpoint, then KL-anchoring to it, lets us target new behaviors without disturbing the ones already in production.

Verifiable rewards, AI feedback, and reward hacking. RL with verifiable rewards optimizes against programmatic checks rather than a learned model (Lambert et al., 2024), while RLAIIF substitutes AI feedback for human labels (Lee et al., 2024), typically via an LLM judge (Zheng et al., 2023). LLM judges are, however, susceptible to reward hacking on attributes such as length and surface diversity. Our reward is deliberately hybrid for this reason: verifiable format and n-gram diversity terms guard exactly the axes a judge scores unreliably, while the judge is reserved for the subjective, non-verifiable quality of contextual adaptation.

3 Method

Our approach grounds creative query suggestions in two stages (Figure 1). Stage 1 builds a grounded suggestion model through corpus-grounded synthetic supervision (§3.1): every suggestion is tied to real assets in the inventory, and one model serves all suggestion types in a single retrieval-free step at low latency. Stage 2 adapts that model with reinforcement learning (§3.2) to meet product requirements that supervision met poorly, namely single-word formats and selective seasonal and regional grounding. Because Stage 2 starts from the grounded model and stays anchored to it, it adds these behaviors without regenerating the grounded data and without changing the serving path.

3.1 Stage 1: Grounded Supervision

Stage 1 is a deployed supervised system we published separately (Jain et al., 2026); we summarize it here as the grounded foundation this paper builds on and refer the reader to that work for full construction and training detail. Its role in this paper

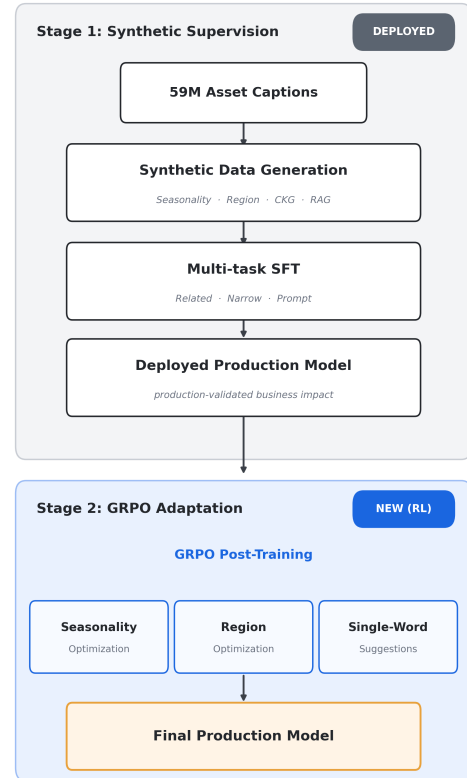


Figure 1: The two stages of this work. Stage 1 is the deployed synthetic-supervision system, validated in production. Stage 2 is the GRPO post-training stage introduced here, which adapts it for selective seasonal and regional grounding and for single-word generation, leaving the retrieval-free serving path unchanged.

is to fix the deployed checkpoint, the retrieval-free serving constraint, and the notion of grounding that Stage 2 must preserve.

The data problem. No suitable training data exists. Log-based query suggestion learns from historical queries and clicks, but on a creative platform these are sparse, noisy, long-tailed, and privacy-constrained; they underrepresent the structured refinements and long generation prompts we need, give no signal for the required context conditioning, and offer no guarantee that a suggestion maps to a real asset. Generating the data synthetically is not straightforward either: a useful example must be conditioned on *asset type*, *seasonality*, and *region* at once and be *grounded* in assets that actually exist, whereas a naive generator applies context indiscriminately and invents fluent but unsupported suggestions that retrieve nothing. Producing supervision that is both correctly conditioned and grounded is the core challenge Stage 1 must solve.

Corpus-grounded synthetic data. We construct supervision synthetically from 59M creative asset captions through a two-stage offline pipeline. The first stage rewrites each sampled caption into a short, intent-faithful seed query and, where caption semantics strongly imply it, attaches one of roughly 170 seasonality tags (region is then derived from a season-to-region mapping and never surfaced in outputs); the second expands each seed query, conditioned on asset type and any inferred context, into related suggestions, narrow refinements, and generation-ready prompts. We sample 500K captions for training examples and use the full 59M corpus for retrieval grounding. Two mechanisms keep the supervision grounded: the *Creative Knowledge Graph* (CKG) constrains how expansions vary across asset-specific dimensions (layout, motif, palette, composition, tone), curbing intent drift and near-synonymy, while *retrieval-augmented grounding* (RAG) supplies the most similar corpus captions as generation context so suggestions correspond to retrievable assets. Both operate only during offline data construction; at serving time the model uses no retrieval.

Model and serving. The generated dataset supervises a single gemma-3-4b-it model, fine-tuned on all three tasks jointly with one shared LoRA adapter under a completion-only objective, pairing a structured instruction (seed query, asset type, seasonality, region) with a JSON-formatted target. For serving, the adapter is merged into the base weights and the model quantized to FP8, giving single-pass, retrieval-free inference at roughly 280 ms. This retrieval-free path is a fixed constraint that any later adaptation must preserve.

Grounding evaluation and production impact. We measure grounding directly rather than by proxy: each suggestion is executed against the production search index, and a judge assesses whether the retrieved assets match the suggested query’s intent, giving practical corpus alignment rather than embedding similarity. During data construction, retrieval-augmented grounding raises the share of well-grounded suggestions (top-bin, judge score ≥ 5) by over 30% relative to ungrounded generation, and the deployed 4B model improves over a smaller variant across all quality dimensions, most sharply on seasonality. In deployment, executing served suggestions against the live index yields a null rate of roughly 1% (about 99% retrieve real assets), our most direct grounding evidence. The

deployed system was also validated against the prior log-based suggester: annotators preferred its suggestions in 72% of paired comparisons versus 11% (the remainder ties) and rated them 4.18/5. These Stage 1 scores use a 0–5 judge scale and are not directly comparable to the 0–100 scale of Stage 2 (§4).

Where supervised grounding falls short.

Strong as it is, the supervised model leaves gaps that surfaced on new surfaces. The supervised data contains only multi-word suggestions, so compact surfaces such as filters, chips, and autocomplete, which need single-word output, are unsupported. Seasonal and regional grounding is a subtler gap. The original evaluation penalized only *spurious* context, added where none applied, and not the *failure* to add context where warranted, so a neutral-by-default model scored well while rarely introducing appropriate context. Production revealed the cost: suggestions carrying appropriate seasonal or regional context perform better, yet the deployed model rarely adds it. The requirement is thus *selective* grounding: add context when relevant, suppress it when not. Regenerating and retraining on new supervision would address this only slowly, and would risk regressing behaviors already in production, so Stage 2 treats it as a reward-design problem.

3.2 Stage 2: Reinforcement Learning for New Requirements

We adapt the deployed model with a GRPO post-training stage that updates only the weights, leaving the data source and serving path unchanged. The hybrid reward is built around two ideas, *contextual selectivity* (apply seasonal and regional context only when it helps) and *verifiable control* (use programmatic checks on the axes a learned judge scores unreliably), together with relevance and ranking terms that preserve existing quality. Anchoring to the grounded Stage 1 model through the KL term (below) is what lets the new behaviors be added without eroding the grounding already in place.

Problem setup. We treat suggestion as conditional generation. A prompt specifies a seed query, asset type, optional seasonal and regional context, the task type (RELATED, NARROW, PROMPT, or the new SINGLE task), and an optional target length band. The policy π_θ emits a structured JSON object whose value is a *set* of suggestions. We initialize π_θ

from the deployed checkpoint, which also serves as the frozen reference π_{ref} for the KL penalty below.

Requirement-aligned evaluation of context.

Before defining the reward, we make the seasonality and region objectives precise, since this is where the supervised model falls short. For each prompt we classify the context into one of three cases and require different behavior in each: **RELEVANT**, where the context applies and should be woven into some of the suggestions; **PARTIAL**, where only the atmosphere is appropriate and the context should be evoked but never named; and **IR-RELEVANT**, where the context does not apply and must be suppressed. For region we add a fourth branch, **ALREADY-SPECIFIED**, for queries that already name a region, where the context is left as given. This classify-then-require formulation turns selective grounding into an explicit, scoreable objective.

GRPO objective. For each prompt we sample a group of G completions from the current policy, score each with the composite reward of §3.2.1, and convert the scores into group-relative advantages by standardizing within the group (Shao et al., 2024). The policy is updated with the clipped objective

$$\mathcal{J}(\theta) = \mathbb{E} \left[\min \left(\rho \hat{A}, \text{clip}(\rho, 1-\epsilon, 1+\epsilon) \hat{A} \right) - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta} \parallel \pi_{\text{ref}}] \right], \quad (1)$$

where ρ is the token-level likelihood ratio between the updated and sampling policies and $\hat{A}$ the group-relative advantage. The KL term anchors the policy to the deployed, grounded model, so the targeted behaviors are not bought at the cost of the relevance, diversity, and grounding already serving production.

3.2.1 Composite Reward

The reward is a weighted sum of the components below. Degenerate outputs (unparseable responses, JSON-schema violations, or the wrong number of items) are discouraged through component-specific penalties and fallback scores rather than a single multiplicative gate, which prevents the policy from exploiting formatting failures while preserving informative gradients; Appendix C gives the exact handling per component. The components follow one design principle: the judge scores subjective quality, while programmatic checks cover the axes the judge can be gamed on.

Format and length (verifiable). A programmatic term rewards the fraction of suggestions whose word count falls in the target band; for the **SINGLE** task the band collapses to exactly one word. Because this signal is dense and exact, it is the primary driver of the new single-word capability and requires no judge call.

Seasonality and region (judge). A judge scores seasonal and regional appropriateness under the policy above: context woven in naturally when **RELEVANT**, atmosphere without explicit labels when **PARTIAL**, and clean suppression when **IR-RELEVANT**. Region is scored separately from seasonality and checks for culturally appropriate motifs and styling rather than any explicit geographic mention. These terms are the primary drivers of selective context application.

Relevance and diversity (judge and programmatic). A relevance term measures intent preservation given the seed query and task, and a set-level diversity term, combining pairwise similarity penalties with distinct-token coverage, discourages near-synonyms. Diversity matters most for the single-word task, where the model could otherwise satisfy the format reward by repeating one frequent token; together with the format penalty it closes that loophole.

Ranking over rollouts (internal). Because the judge scores subjective quality, the G rollouts for a prompt often receive bunched, near-equal composite scores; the resulting group relative advantages collapse toward zero and the policy receives little gradient. We therefore add a term that converts the judge’s assessment into a rank over the group: the judge orders the G rollouts from best to worst and each is scored by its rank position (Appendix C), replacing an easily-compressed set of absolute scores with evenly separated ones so a stable advantage remains. The term is a rank transform of the same judge-quality signal, not a separate objective, so we do not evaluate it standalone.

Weights. Component weights are set on a validation split to balance selective context application and format control against the relevance and diversity we wish to preserve; per-task weights are in Appendix C.

3.2.2 Eliciting the Single-Word Capability

The **SINGLE** task is added to the prompt pool with a one-word target, even though no single-word ex-

amples exist in the supervised data. Because the format reward is verifiable and dense across the sampled group, GRPO can surface this latent capability without new demonstrations: the model already knows the words; it simply never learned to emit them alone. To keep the new task from crowding out the others, we mix it with the three existing tasks at a fixed sampling ratio chosen on validation data.

3.2.3 Why Reinforcement Learning Here

Both target behaviors are easier to reward than to demonstrate: a single-word constraint is a verifiable check and selective context fits an asymmetric reward, whereas teaching either by example would require regenerating the data pipeline (with no single-word targets to seed from) and retraining, at the risk of regressing what already works. GRPO optimizes these requirements directly on the deployed model without touching the retrieval-free serving path.

4 Experimental Setup

Tasks and data. We evaluate the Stage 2 adaptation on three held-out test sets, all decoded at $T=0.7$ to match deployment: seasonality (1,544 prompts), region (1,125), and related search (500). The three share the reward architecture but differ in routing and output target (Table 1). Seasonality uses the three-case policy of §3.2 (RELEVANT, PARTIAL, IRRELEVANT); region adds an ALREADY-SPECIFIED branch for queries that already name a region; and related search applies no routing, targeting a one-word band against the 6–8-word band of the other two.

Models and baselines. The policy is gemma-3-4b-it with the Stage 1 SFT LoRA merged into the base, then GRPO-trained with a fresh low-rank adapter ($r=16$) via TRL. Our baseline is the deployed Stage 1 SFT checkpoint (§3.1); comparing against it isolates the effect of the reinforcement-learning stage. For related search we additionally evaluate the same checkpoint prompted directly for one-word output, to separate the GRPO gain from instruction following alone. The reward judge is Qwen3.5-27B (Qwen Team, 2026), an instruction-tuned model from a different family than the Gemma policy, so the judge never scores its own family. Full hyperparameters and per-task reward weights are in Appendix C; the judge and classification

	Seas.	Region	RS
Test prompts	1,544	1,125	500
Eval temp.	0.7	0.7	0.7
Length band	6–8	6–8	1
Judge routing	3-way	4-way	none

Table 1: Per-task evaluation setup. Region routing is the three-case policy plus an ALREADY-SPECIFIED branch. Reward weights and full hyperparameters are in Appendix C.

prompts are in Appendix D.

Metrics. Our primary metric is judge quality on a 0–100 scale: the class-routed geometric mean of the rubric dimensions (§3.2), reported overall and per judge-assigned class. This is a requirement-aligned metric that penalizes both spurious and missing context, and it therefore differs from, and is not directly comparable to, the 0–5 Stage 1 evaluation of §3.1, which did not penalize missing context. Programmatic checks (format compliance, length-band rate, n-gram diversity) are reported as diagnostics. For the related-search parity comparison we report 95% confidence intervals from a paired bootstrap ($B=10,000$ resamples over the 500 prompts). We exclude ranking from standalone evaluation: it enters only as an internal reward component (§3.2.1) and degenerates to a constant at single-rollout ($G=1$) evaluation.

Class assignment. Each test prompt is routed to a context class by the judge from the input alone, independently of the model under evaluation. The class mix is therefore identical for the baseline and GRPO conditions—seasonality 250/710/584 (16.2/46.0/37.8%) over 1,544 prompts, and region 407/357/257/104 (36.2/31.7/22.8/9.2%, REL/PART/IRR/SPEC) over 1,125—so every per-class row compares the two models on the same prompts, and the overall score is the proportion-weighted mean of the per-class composites.

5 Results

We report each requirement in turn: selective seasonal grounding (the primary result), the same framework transferred to region, and single-word controllability. Throughout, the baseline is the deployed Stage 1 checkpoint, so every gain is attributable to the GRPO stage alone.

Seasonality (primary result). The per-class breakdown (Table 2) exposes the failure mode and

Adapter	Overall	REL.	PART.	IRR.
Baseline	53.9	32.9	30.4	91.5
GRPO	72.5	62.5	59.6	92.5

Table 2: Seasonality judge quality by judge-assigned class (1,544 prompts, $T=0.7$). Overall improves from 53.9 to 72.5 (+18.6 points). GRPO improves all three classes at once, with the largest gains on the two hard classes and IRRELEVANT held at ceiling.

Adapter	Overall	REL.	PART.	IRR.	SPEC.
Baseline	54.1	39.4	35.0	95.7	73.6
GRPO	59.2	47.8	40.7	96.3	75.9

Table 3: Region judge quality by category (1,125 prompts, $T=0.7$). Overall improves from 54.1 to 59.2 (+5.2 points). SPEC. denotes ALREADY-SPECIFIED (class sizes 407/357/257/104). GRPO lifts the two hard categories (REL., PART.) while holding IRRELEVANT and ALREADY-SPECIFIED at ceiling, mirroring the seasonality result.

the fix. The SFT baseline is near-ceiling on IRRELEVANT queries, since suppressing context when the query carries none is low-difficulty, but it fails on both contextually hard classes. GRPO improves both hard classes substantially while holding IRRELEVANT at ceiling: all three classes improve at once, with no trade-off. The PARTIAL class, the largest of the three at roughly 46% of the 1,544 test prompts, drives the bulk of the +18.6-point overall gain.

Region (generalization). We apply the identical framework to a region axis, changing only the training data and judge prompts and adding the ALREADY-SPECIFIED branch. Table 3 shows the same pattern: the two hard categories improve, IRRELEVANT and ALREADY-SPECIFIED stay at ceiling, and no category regresses. The classify-then-route reward transfers to a second contextual axis with no changes to the model or training setup, only an added routing branch and a task-specific judge prompt.

Related search (lexical controllability). The SFT model cannot reliably follow a single-word instruction: its prior toward multi-word phrases means prompting alone yields a true single word only 15.1% of the time (500 test prompts). GRPO raises this to 99.1%, unlocking a format absent from the supervised distribution. This gain comes at no cost in the normal regime: under a standard 3–5-word instruction, GRPO is statistically indis-

tinguishable from the baseline on both judge dimensions (Table 4). A paired bootstrap over the 500 prompts ($B=10,000$) puts the GRPO–baseline difference at +0.7 points on both relevance (95% CI $[-0.2, 1.7]$) and diversity (95% CI $[-0.3, 1.7]$); both intervals straddle zero, so the apparent gains are within sampling noise. The single-word dips in diversity and relevance are expected costs of compressing a query to one word, and the baseline only avoids them by ignoring the instruction and returning multi-word phrases most of the time. Appendix E gives qualitative examples;

Grounding preservation after GRPO. Because the RL stage leaves the serving path unchanged, we execute the GRPO policy’s outputs against the production index, as in Stage 1. Coverage stays at or near ceiling (100.0% seasonality and region, 99.8% related search, 99.5% single-word), matching the deployed system’s near-zero null rate, and retrieved assets are relevant for 89.7%, 88.6%, and 83.1% of multi-word suggestions and 72.9% of single-word suggestions, the expected cost of one-token compression.

Human preference study. To validate the Stage 2 gains beyond the LLM judge, we ran a blinded pairwise study on the two context axes the GRPO stage targets: GRPO-vs-baseline suggestion pairs, each judged by two of ten annotators against a hidden A/B key, recorded as GRPO, baseline, or tie. Inter-annotator agreement is fair (Fleiss’ $\kappa = 0.34$). On decided pairs (ties excluded), humans prefer GRPO on 59% of seasonality pairs and 55% of region pairs, a real if modest edge on both axes, consistent with the automated judge results. On the single-word task, GRPO is preferred in roughly 95% of comparisons, since the SFT baseline rarely produces single-word output; the rare cases where it does are unsuitable for the dynamic filter.

6 Conclusion

We presented a creative query-suggestion system grounded in two stages. Stage 1 establishes grounded suggestions through corpus-grounded synthetic supervision: with no user logs, it ties suggestions to real assets, serves them in a single retrieval-free step at roughly 280 ms, and is validated in production, where served suggestions return real assets at a null rate of about 1%. Stage 2 adapts this deployed model with a GRPO post-

Model	Cond.	1-wd	Rel.	Div.
Baseline	1-wd	15.1	84.5	75.1
GRPO	1-wd	99.1	72.8	52.3
Baseline	3–5	0.0	86.6	83.4
GRPO	3–5	0.0	87.3	84.1

Table 4: Related search (500 prompts, $T=0.7$). “1-wd” is the single-word rate; judge dimensions are on the 0–100 scale. Under the normal 3–5-word instruction the two models are statistically indistinguishable on both judge dimensions (paired bootstrap, $B=10,000$: relevance $\Delta=+0.7$, 95% CI $[-0.2, 1.7]$; diversity $\Delta=+0.7$, 95% CI $[-0.3, 1.7]$); the single-word dip is specific to the one-word extreme, not a general regression.. The “3–5” condition is the unconstrained related-search regime (no single-word target imposed) and serves as a control; it is distinct from the 6–8-word band used for the seasonality and region evaluations (Table 1).

training stage to meet requirements that supervised data transferred poorly: selective seasonal and regional grounding, and reliable single-word suggestions. A hybrid reward pairing verifiable format and length signals with judge-based quality improved seasonality by 18.6 and region by 5.2 points and lifted single-word compliance from 15.1% to 99.1%, with relevance and diversity preserved in the normal multi-word regime and no change to the retrieval-free serving path. Because the stage starts from the grounded model and stays anchored to it, these behaviors are added without regenerating the grounded data or disturbing what already serves production. More broadly, the result suggests that reinforcement learning is an effective, low-risk complement to synthetic supervision for adapting deployed, grounded systems to behaviors that are easier to *score* than to *demonstrate*.

Limitations

Our reward is partly model-based, so judge bias may transfer into the policy. The training reward and the primary evaluation share the same judge family; using a fully independent, different-family judge for evaluation remains future work. We also do not report a controlled ablation of the individual reward components; their contributions are argued from development observations (for example, the n-gram diversity term was added after outputs collapsed onto a repeated template), and isolating each term quantitatively is left to future work. Our evaluation is automated throughout: while the deployed SFT system was validated with human preference and live business metrics, the GRPO gains are so

far confirmed by LLM judges and informal author inspection, not yet by a controlled human study or a live click-through test, so the production effect of the improved seasonal, regional, and single-word behaviors is not yet measured online. Finally, GRPO adds offline training compute, though inference is unchanged.

Ethics Statement

The system is trained on synthetic, corpus-derived supervision and avoids user search logs, reducing exposure of behavioral and personal data. Regional adaptation is expressed through culturally appropriate motifs and never surfaces explicit geographic labels. To mitigate cultural stereotyping in seasonal and regional outputs, suggestions are constrained by the Creative Knowledge Graph and grounded in the asset corpus, and the judge rubric penalizes outputs that rely on reductive or caricatured cultural cues. We used AI-based writing assistance to draft and edit portions of this paper; all technical content, experiments, and claims are the authors’ own.

References

- Jinheon Baek, Nirupama Chandrasekaran, Silviu Cucerzan, Allen Herring, and Sujay Kumar Jauhar. 2024. Knowledge-augmented large language models for personalized contextual query suggestion. In *Proceedings of the ACM Web Conference (WWW)*.
- Ricardo Baeza-Yates, Carlos Hurtado, and Marcelo Mendoza. 2004. [Query recommendation using query logs in search engines](#). In *Current Trends in Database Technology – EDBT 2004 Workshops*, pages 588–596. Springer.
- Luiz Bonifacio, Hugo Abonizio, Marzieh Fadaee, and Rodrigo Nogueira. 2022. InPars: Unsupervised dataset generation for information retrieval. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 2387–2392.
- Zhuyun Dai, Vincent Y. Zhao, Ji Ma, Yi Luan, Jianmo Ni, Jing Lu, Anton Bakalov, Kelvin Guu, Keith B. Hall, and Ming-Wei Chang. 2023. Promptagator: Few-shot dense retrieval from 8 examples. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*.
- DeepSeek-AI. 2025. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*.
- Harshit Jain, Twisha Naik, Vipul Jain, Chirag Arora, Sudhakar Pandey, Vedanth Subramaniam, and Es-

- han Trivedi. 2026. Efficient context-aware corpus-grounded query and prompt suggestion constrained by creative knowledge graph. In *Workshop on Synthetic Data Generation and Management for Building AI Systems*, pages 27–37.
- Vitor Jeronimo, Luiz Bonifacio, Hugo Abonizio, Marzieh Fadaee, Roberto Lotufo, Jakub Zavrel, and Rodrigo Nogueira. 2023. InPars-v2: Large language models as efficient dataset generators for information retrieval. *arXiv preprint arXiv:2301.01820*.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. 2023. [Survey of hallucination in natural language generation](#). *ACM Computing Surveys*, 55(12):1–38.
- Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, and 1 others. 2024. Tulu 3: Pushing frontiers in open language model post-training. *arXiv preprint arXiv:2411.15124*.
- Harrison Lee, Samrat Phatale, Hassan Mansoor, Thomas Mesnard, Johan Ferret, Kellie Lu, Colton Bishop, Ethan Hall, Victor Carbune, Abhinav Rastogi, and Sushant Prakash. 2024. RLAIIF vs. RLHF: Scaling reinforcement learning from human feedback with AI feedback. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 9459–9474.
- Erxue Min, Hsiu-Yuan Huang, Xihong Yang, Min Yang, Xin Jia, Yunfang Wu, Hengyi Cai, Junfeng Wang, Shuaiqiang Wang, and Dawei Yin. 2025. From prompting to alignment: A generative framework for query recommendation. *arXiv preprint arXiv:2504.10208*.
- Rodrigo Nogueira, Wei Yang, Jimmy Lin, and Kyunghyun Cho. 2019. Document expansion by query prediction. *arXiv preprint arXiv:1904.08375*.
- Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. 2024. Nomic embed: Training a reproducible long context text embedder. *arXiv preprint arXiv:2402.01613*.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, and 1 others. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Qwen Team. 2026. [Qwen3.5: Towards native multi-modal agents](#).
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Yu Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.
- Nandan Thakur, Jianmo Ni, Gustavo Hernández Ábrego, John Wieting, Jimmy Lin, and Daniel Cer. 2024. Leveraging LLMs for synthesizing training data across many languages in multilingual dense retrieval. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

A Stage 1 Training and Serving Configuration

Corpus and sampling. The corpus is ~59M vision-language captions of platform assets (images, videos, templates, backgrounds, icons). Captions are cleaned (dropping empty, malformed, or >30-word entries and profanity), embedded with Nomic ([Nussbaum et al., 2024](#)), and indexed for retrieval. We sample 500K captions to generate training examples; the full corpus is used for grounding.

Data generation. Stage 1 rewrites each caption into a seed query and infers seasonality (~170 tags, assigned only when strongly implied); region follows a season-to-region mapping. Stage 2 expands each seed query, conditioned on asset type, seasonality, and region, into 10–15 related suggestions, 5–10 refinements, and 3–5 generation prompts. Creative Knowledge Graph dimensions (layout, palette, motif, composition, tone) constrain how outputs vary, and retrieval-augmented grounding supplies the most similar captions as context. Negative seasonality is excluded from generation and handled later by the Stage 2 reward.

Fine-tuning. We fine-tune gemma-3-4b-it on all three tasks jointly with one shared LoRA adapter and completion-only loss, over chat-style instruction–response pairs with ~ 20 instruction variants per task. LoRA: $r=16$, $\alpha=32$, dropout 0.2, all attention and MLP projections. Optimization: 15 epochs, lr 4×10^{-5} , weight decay 0.05, linear schedule with 10% warmup, batch size 4, gradient accumulation 4, max length 1536, bf16. The best checkpoint is chosen by validation loss and judge metrics.

Serving. The adapter is merged and the model quantized to FP8 for single-step, retrieval-free inference at ~ 280 ms. The multi-step generation pipeline runs entirely offline.

Evaluation. Offline evaluation uses ~ 30 K held-out queries stratified by frequency (head, torso, tail), month, and region. An LLM judge scores relevance, diversity, seasonality, region, and grounding consistency; grounding consistency executes each suggestion against the production index and judges whether the retrieved assets match intent.

B Stage 1 Ablations

Retrieval-augmented grounding. The main-body ablation (Table ??) compares expanding seed queries with and without retrieved captions as grounding context. RAG raises the share of well-grounded suggestions (top-bin, score ≥ 5) by over 30% relative and improves relevance, diversity, and seasonality.

Creative Knowledge Graph. Without CKG constraints, generation clusters on near-synonyms and drifts to generic reformulations (for “pumpkin poster”: “pumpkin flyer”, “pumpkin design”, “autumn poster”), sometimes adding attributes unsupported by the asset type. CKG-guided variation across layout, motif, palette, composition, and tone yields distinct interpretations (“rustic pumpkin harvest poster”, “minimalist autumn flyer with warm palette”) and reduces drift, improving controlled diversity without hurting relevance.

LoRA rank. Rank 8 slightly reduced diversity and seasonality; rank 32 gave marginal relevance gains but more overfitting and instability. Rank 16 was the best balance.

Component	Seasonal	Region	RS
w_{rank}	0.45	0.40	0.20
w_{judge}	0.30	0.40	0.30
w_{format}	0.05	0.05	0.10
w_{length}	0.05	0.05	0.30
w_{ngram}	0.15	0.10	0.10

Table 5: Per-task composite reward weights (sum to 1.0).

C Reward and Training Details

Composite reward weights. Table 5 lists the per-task weights. The judge term w_{judge} covers the seasonality/region and relevance dimensions scored together by the judge (below), and w_{ngram} is the programmatic diversity term of §3.2. RS raises w_{length} to 0.30 because length compliance is the main training signal; region raises w_{judge} to 0.40 because cultural nuance is not programmatically verifiable. Although w_{rank} is the largest single weight for the seasonality and region tasks, the ranking term is not an independent objective: it rank-transforms the judge’s own quality assessment across the G roll-outs (§C, “Ranking reward”) to keep group-relative advantages from collapsing when the subjective scores bunch together. It carries the largest weight precisely because this dispersion is what makes the judge signal trainable under GRPO, not because it optimizes a different target. Being a monotone transform of judge quality, it is reported through that metric: we use judge quality, not ranking, as the standalone number (§4).

Parse-failure handling. Each component handles parse failure independently: `format_fn` and `length_fn` return 0.0 on an empty suggestion list; `ngram_diversity_fn` returns 50.0 (neutral) when fewer than two suggestions parse; the judge falls back to 30.0 per dimension on any API or parse failure. Together these push degenerate outputs down.

Format reward. Starts at 100 per suggestion, with penalties: -20 trailing punctuation (`. ! ? :`), -15 double spaces, -40 underscores, -30 for two or more hyphens (the last two close tokenizer concatenation hacks, in the region and RS implementations). The per-suggestion average is then penalized at the group level: -30 if the count is off by one, -60 if off by two or more.

Length reward. Word count maps to 100 (in-band), 70 (± 1), 40 (± 2), 10 (further). For SINGLE

Dimension	REL.	PART.	IRREL.
season_quality	0.60	0.45	0.30
season_balance	0.25	0.30	0.50
relevance	0.15	0.25	0.20

Table 6: Judge dimension weights by classification (seasonality). For IRREL., season_balance (target of zero seasonal suggestions) dominates; for PART., season_quality rewards atmospheric capture without naming the season.

the band is exactly one word, and a no-separator string over 14 characters is treated as a concatenation and given the floor score. The group score is the suggestion-level mean.

N-gram diversity reward. Pairwise bigram Jaccard overlap over all $\binom{N}{2}$ pairs in the group (bigrams lowercased, punctuation-stripped); the reward is $100 \times (1 - \bar{J})$ for mean Jaccard $\bar{J}$. We added this after early runs collapsed onto a “vibrant/colorful + diverse [X]” template the judge scored highly; programmatic Jaccard cannot be gamed the same way.

Ranking reward. The judge ranks all G rollouts for a prompt, ordering them by assessed quality. Rank position k (0-indexed, best = 0) scores

$$R_{\text{rank}}(k) = 100 \times \frac{G - 1 - k}{G - 1},$$

giving 100 to the best rollout and 0 to the worst; on failure the order is shuffled to avoid systematic bias.

Judge quality composite. For seasonality, the judge scores three dimensions (0–100): season_quality, season_balance, and relevance; region uses an analogous set with region appropriateness in place of the seasonal dimensions. The composite is a weighted geometric mean, with weights routed by the judge’s own classification of the prompt (Table 6). Classification runs at temperature 0.0 (≤ 80 tokens); scoring at temperature 0.1 (up to 1024 tokens for seasonal and region, 512 for RS). On failure all dimensions fall back to 30.0.

GRPO hyperparameters. Table 7 lists the settings shared across tasks. All three share the merged SFT base checkpoint and train a fresh LoRA adapter on top.

Hyperparameter	Value
Group size G	8
KL coefficient β	0.05
PPO clip ϵ	0.2
Advantage norm.	per-prompt z -norm
Learning rate	5×10^{-5}
LR scheduler	cosine, 5% warmup
Batch size	1 (per device)
Gradient accumulation	8
Max gradient norm	1.0
Rollout temperature	1.2
Rollout top- p	0.95
Max prompt length	2048
Max completion length	512 (seas.) / 256 (region, RS)
LoRA r / α / dropout	16 / 32 / 0.05
LoRA targets	all-linear
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)

Table 7: GRPO hyperparameters shared across tasks. Per-task budgets: seasonal max_steps=2400 (patience 5), region 1200 (patience 5), RS 600 (patience 3). Rollout temperature 1.2 is for training sampling; all evaluation uses $T=0.7$.

D Judge and Classification Prompts

Judge model. All judge and classification calls use Qwen3.5-72B (Qwen Team, 2026), a dense instruction-tuned model from a different family than the gemma-3-4b-it policy. This separation is deliberate: the judge never evaluates outputs from its own model family, which mitigates the self-preference bias to which same-family evaluation is prone.

Two-step scoring. Seasonality and region are scored in two steps. A classification call first routes each prompt to a context class from the input alone—the seed query, asset type, and the seasonal or regional context provided—at temperature 0.0; a scoring call then applies the rubric for that class at temperature 0.1. Seasonality routes three ways (RELEVANT/PARTIAL/IRRELEVANT); region routes four, adding ALREADY-SPECIFIED for queries that themselves name a region, where the model must honor the embedded region and ignore the separate region field. Related search uses no routing: every prompt is scored by a single rubric. Because classification depends only on the input, the class assignment is identical across the models under comparison (§4).

Scoring rubrics. Each seasonality and region scoring call returns a short free-text analysis followed by a JSON object of three 0–100 dimensions. For seasonality these are season_quality (depth of seasonal integration in the best sugges-

tion), season_balance (whether context is applied to the right number of suggestions rather than all or none), and relevance; region uses the analogous set with region appropriateness in place of the seasonal dimensions. The class-routed composite (§3.2.1, Table 6) is a weighted geometric mean over these three. Set-level diversity on the seasonality and region tasks is not judged but guarded programmatically by the n-gram term (§3.2.1), which is harder to game than a judged score. On any API or parse failure a dimension falls back to a neutral 30.0.

Representative prompts. We reproduce two seasonality prompts: the classification prompt (Listing 1), which routes each input to a context class, and the RELEVANT scoring rubric (Listing 2). The remaining prompts follow the same structure with class- and task-specific substitutions: PARTIAL replaces the quality and balance dimensions with their atmosphere-without-naming variants (the season must be evoked but never stated), IRRELEVANT rewards clean suppression of context, region uses the analogous rubric with a fourth ALREADY-SPECIFIED branch, and related search applies a single unrouted rubric with a naturalness dimension that floors concatenated or separator-joined tokens (e.g. gaming_poster). The listings are lightly condensed for typesetting: three Unicode glyphs (arrows, em-dashes, curly quotes) are normalized to ASCII, and the classification prompt is shown with abbreviated category definitions and a representative subset of its per-class few-shot examples.

Listing 1: Seasonality classification prompt (category definitions and few-shot examples abbreviated). The scoring stage applies the rubric for the returned class.

```
You are a search product manager at a stock asset platform.
Decide:
if a real user submitted this query during this season,
would
seasonally-influenced suggestions actually help them?

RELEVANT -- The query is thematic or occasion-driven;
seasonal
suggestions directly serve the user's intent.
    "festive food photography" + Diwali -> user making Diwali
    food
        content; seasonal is the point
    "table setting" + Thanksgiving -> holiday table settings
    are
        exactly what this user needs
    "wedding photography" + Spring -> spring weddings are high
    -demand;
        seasonal framing helps

PARTIAL -- The query is a generic asset type; seasonal mood
could
inspire one atmospheric suggestion, but the season must
never be
named directly.
    "abstract background" + Winter -> one cool-toned
    atmospheric
```

```
variant has value; "winter background" does not
"seamless pattern" + Easter -> pastel spring motifs serve
some
users; forcing "Easter pattern" does not

IRRELEVANT -- The query is specific/utilitarian, the season
is
plainly mismatched with the asset's purpose, or the seed
already
names the season.
    "coffee cup" + Diwali -> user wants a coffee cup; Diwali
    framing
        produces worse results
    "professional background" + Easter -> business asset;
    Easter
        theming is inappropriate for the use case
    "Halloween flyer" + Halloween -> seed already names the
    event;
        seasonal reinforcement is redundant

Output ONLY a JSON object (no markdown fences):
{"classification": "RELEVANT" | "PARTIAL" | "IRRELEVANT"}
```

Listing 2: Seasonality RELEVANT scoring rubric. PARTIAL and IRRELEVANT share this structure with class-specific dimension definitions.

```
You are a precise evaluator of search query suggestion
GROUPS. The
seasonality field is RELEVANT to the seed query -- score the
group
accordingly.

You will receive the user's search intent, contextual
parameters
(asset type, region, seasonality), and a group of generated
suggestions. Each suggestion is a short search query of
roughly
4-8 words.

Output:
1. Write 2-3 sentences inside <analysis> tags.
2. Then output a single JSON object with three float scores
(0.0 to 100.0). No markdown fences.

CRITICAL SCORING RULE -- YOU MUST USE THE FULL RANGE:
- Poor group (wrong season, wrong format, incoherent): 5-25
- Mediocre group (technically correct but weak/generic):
25-45
- Average group (acceptable but unremarkable): 45-60
- Good group (natural, relevant, well-balanced): 65-80
- Excellent group (evocative, specific, perfect balance):
85-100
Scoring 3+ consecutive groups between 50-70 on any single
dimension
means you are not discriminating. Force yourself to use the
extremes.

DIMENSION 1: season_quality (0.0-100.0) -- Quality of
seasonal
integration. Evaluate the BEST seasonal suggestion in the
group. How
naturally and deeply does it incorporate the seasonal
context?
    0-10: No suggestion attempts seasonal integration despite
    relevance.
    10-25: Season/event name mechanically appended -- no
    thematic depth.
    25-40: Season mentioned as a label but not woven into
    query meaning.
    40-60: Acceptable thematic connection between season and
    query.
    60-80: Seasonal themes naturally woven into a specific
    search query.
    80-100: Evocative language capturing the spirit of the
    season.
Key distinction: "outdoor grilling patriotic table" (
thematic) >
    "Independence Day cooking" (label appending).

DIMENSION 2: season_balance (0.0-100.0) -- Is seasonality
applied to
the right number of suggestions? First, write "Seasonal
count: X/3"
(count suggestions with seasonal context). Then score:
```

```

0-10: 0 out of 3 -- model completely ignored relevant seasonality.
10-25: 3 out of 3 -- model over-applied, no non-seasonal diversity.
25-45: 2 out of 3 -- too much seasonal content, not enough variety.
45-65: 1 out of 3 but the seasonal suggestion is weak/superficial.
65-85: 1 out of 3 with decent seasonal integration.
85-100: Exactly 1 out of 3 with strong, natural integration -- ideal.

DIMENSION 3: relevance (0.0-100.0) -- Are all suggestions relevant to the seed query and asset type?
0-15: Most suggestions are irrelevant or incoherent.
15-30: Tangentially related, miss the core intent.
30-50: Somewhat related but vague or awkward.
50-70: Relevant but generic.
70-85: Natural search queries addressing the user's intent well.
85-100: Specific, natural queries perfectly matching intent and type.

RESPONSE FORMAT (you MUST follow this exactly):
<analysis>Your 2-3 sentence analysis of the group.</analysis>
{
  "season_quality": XX.X,
  "season_balance": XX.X,
  "relevance": XX.X
}

```

E Qualitative Examples

Each example shows a seed query and context, the SFT baseline suggestions, and the GRPO output. Suggestions that exhibit the target behaviour are **bolded**.

(a) Seasonality: “floral arrangement” + Lunar New Year (RELEVANT). Lunar New Year floral arrangements are a real search category, so the seasonal context is RELEVANT and should be woven into at least one suggestion. SFT ignores the context entirely; GRPO introduces the canonical LNY palette (red and gold), auspicious peonies, and cherry blossom associations while keeping one neutral suggestion for diversity.

(b) Region: “crowd gathering” + India (RELEVANT). Region India provides strong visual cues (festival attire, market settings, fireworks) that make suggestions more useful. The ideal output grounds suggestions in those motifs without an explicit geographic label. SFT returns interchangeable generic queries; GRPO introduces culturally specific imagery.

(c) Single-word: “new year party” (RELATED SEARCH). The query is issued with a one-word length instruction. SFT’s prior pulls toward multi-word phrases regardless of the constraint. GRPO, trained with a verifiable length reward, emits single tokens that are high-recall search terms for the topic.

Query	“floral arrangement”
Season	Lunar New Year
Model	Suggestions
SFT	colorful floral arrangement with vase on wooden table romantic outdoor wedding floral arrangement with pastel flowers modern minimalist floral arrangement with greenery and flowers
GRPO	luxury red and gold floral arrangement with peonies vibrant floral wreath with cherry blossoms and greenery modern minimalist floral centerpiece for dining table

Table 8: Seasonality RELEVANT example. SFT produces three generic queries with no seasonal signal. GRPO’s first two suggestions are grounded in Lunar New Year (red/gold palette, peonies, cherry blossoms); the third stays neutral, preserving diversity.

Query	“crowd gathering”
Region	India
Model	Suggestions
SFT	large crowd gathering outdoors in festive atmosphere diverse crowd of people gathered at event crowd gathering in a public square with banners
GRPO	vibrant crowd gathering at night with fireworks diverse crowd in traditional attire at festival large crowd gathering in a bustling Indian market

Table 9: Region RELEVANT example. SFT suggestions could apply to any country. GRPO grounds two suggestions in visual signatures associated with India (festival attire, bustling market) through imagery rather than an explicit geographic mention.

Query	“new year party”
Model	Suggestions
SFT	midnight celebration champagne toast
GRPO	confetti champagne

Table 10: Single-word format example. SFT emits two-word phrases despite the one-word instruction, consistent with its 15.1% compliance rate. GRPO’s outputs are compliant and grounded: *confetti* and *champagne* are core visual search terms for new year party imagery. System-level compliance: SFT 15.1% → GRPO 99.1% (500-prompt eval).